Euclidean Geometry In Mathematical Olympiads

Lemma 1.42: Common Centers

Week 1 | geometry | incenter | Chapter 1 | fgm

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Mar 12, 2018, 4:30 PM • 3 #1

Lemma 1.42
Let ABC be an acute triangle inscribed in circle ω. Let X be the midpoint of the arc BC not containing A and define Y, Z similarly. Show that the orthocenter of $\triangle XYZ$ is the incenter I of $\triangle ABC$

This post has been edited 1 time. Last edited by kootrapali, Aug 13, 2020, 9:35 AM

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Solution

It is known by the incenter/excenter lemma that $A, I,$ and X are collinear. Denote arc $BC = 2\alpha, AB = 2\beta, AC = 2\gamma$.

By the inscribed angle theorem $\angle YZX = \dfrac{\alpha+\beta}{2}$, and $\angle AZX = \dfrac{\gamma}{2}$. By the triangle sum, observe that $YZ \perp XA$. Thus, by symmetry, the result follows.

Vrangr
1600 posts

Mar 23, 2018, 6:38 AM • 4 #4

Second Proof

Note that the incentre of the orthic triangle of an acute triangle is the orthocenter of the reference triangle.

Now reflect I in X, Y, Z to get the excentres, I_A, I_B, I_C. Note that ABC is the orthic triangle of $I_AI_BI_C$, since $IBI_A = \frac{1}{2}(180 - \angle B) + \frac{1}{2}\angle B = 90^\circ$ and other similar results..

Now take homothety with centre I and ratio $\frac{1}{2}$.

This post has been edited 2 times. Last edited by Vrangr, Mar 23, 2018, 6:39 AM

Kagebaka
3001 posts

Sep 30, 2018, 7:53 PM • 3 #5

> *realquarterb wrote:*
> Solution

I don't understand this solution 😕

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chfrn
66 posts

May 26, 2019, 11:59 AM • 2 #6

Alternate

Can also prove A, I, X collinear by showing $\angle IAC = \angle XAC$. Because angle bisector, $\angle IAC = \dfrac{\angle A}{2}$, and because X is midpoint of arc BC, it follows that $\angle XAC = \dfrac{\angle A}{2}$.

Rest of proof same as realquarterb's.

Edit: in realquarterb's proof, I think there was a labeling error.

Potential fix

> *Quote:*
>
> It is known by the incenter/excenter lemma that $A, I,$ and X are collinear. Denote arc $BC = 2\alpha, AB = 2\beta, AC = 2\gamma$.

$$\angle XZY = \frac{a+\gamma}{2}, \angle AXZ = \frac{b}{2}.$$

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